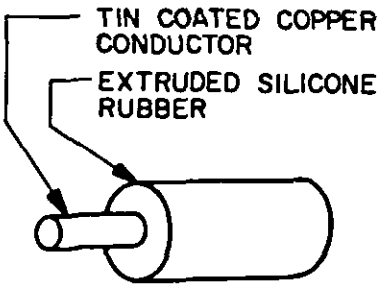
MIL-W-16878/8A(NAVY)
10 August 1981
SUPERSEDING
MIL-W-16878/8(NAVY)(IN PART)
16 January 1961, and
MIL-W-16878/9A(NAVY)(IN PART)
11 May 1963
(See supersession data herein)

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, SILICONE RUBBER INSULATED, 200°C, 1000 VOLTS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878.



SH11877

FIGURE 1. Construction.

TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire Diameter (inch)	
		Material	Coating		Min	Max
24	7 X 32	Copper	Silver	0.024	0.083	0.091
22	7 X 30	Copper	Silver	.030	.089	.097
20	7 X 28	Copper	Silver	.038	.097	.105

FSC 6145

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TABLE I. Construction details. - Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire Diameter (inch)	
		Material	Coating		Min	Max
20	10 X 30	Copper	Silver	0.038	0.097	0.105
18	7 X 26	Copper	Silver	.048	.107	.115
18	16 X 30	Copper	Silver	.048	.107	.115
16	19 X 29	Copper	Silver	.057	.116	.124
16	26 X 30	Copper	Silver	.057	.116	.124
14	19 X 27	Copper	Silver	.072	.161	.171
14	41 X 30	Copper	Silver	.072	.161	.171
12	19 X 25	Copper	Silver	.091	.180	.190
12	65 X 30	Copper	Silver	.091	.180	.190
10	37 X 26	Copper	Silver	.111	.200	.210
10	105 X 30	Copper	Silver	.111	.200	.210
8	133 X 29	Copper	Silver	.169	.281	.307
6	133 X 27	Copper	Silver	.213	.326	.351
4	133 X 25	Copper	Silver	.269	.382	.407
2	133 X 23	Copper	Silver	.335	.448	.473
1	259 X 25	Copper	Silver	.378	.519	.550
0	259 X 24	Copper	Silver	.424	.565	.596
00	259 X 23	Copper	Silver	.477	.618	.649
000	259 X 22	Copper	Silver	.533	.694	.725
0000	259 X 21	Copper	Silver	.601	.742	.773

REQUIREMENTS:

Visual and mechanical inspection: Applicable.
Spark test: 5.0 kilovolts (kV).
Impulse dielectric test: 8 kV.
Tank test: 3.0 kV.
Dielectric withstanding voltage: 3.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C
K = 4000
D = Maximum average diameter of finished wire
d = Conductor diameter

Cold bend: Condition 4 hours at minus 55°C ± 1°C (see table II).

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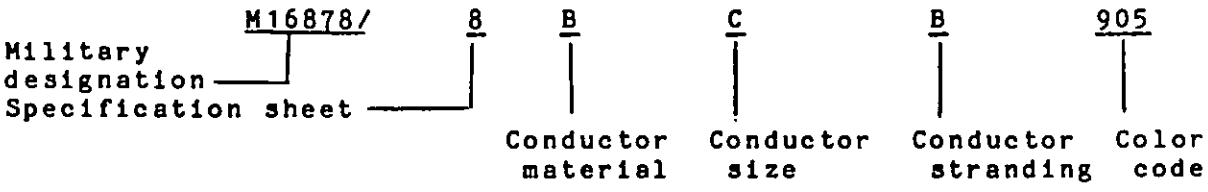
TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter) (inches)
24, 22	1
20 through 12	2
10 through 6	3
4, 2	4.5
1, 0	6
00, 000, 0000	10

Concentricity: 70 percent minimum.
Surface resistance: 5 megohms (minimum).
Wrap back: Not applicable.
Heat resistance: Condition at 250°C.
Flammability: Applicable - 30 seconds (maximum) burning time
3 inches (maximum) flame travel
Soldering: Applicable.
Heat aging: Not applicable.
Insulation tensile strength: 700 pounds force per square inch (minimum).
Insulation elongation: 125 percent (minimum).
Stripe durability: Applicable.

Fungus resistance: Applicable.
Maximum direct current resistance to finished wire: See table I of MIL-W-16878.

Military part number (see MIL-W-16878):



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Supersession data: The wire covered by this specification sheet supersedes MIL-W-16878/8(NAVY), type FF, and MIL-W-16878/9A(NAVY), type FFW, without covering or jacket.

eview activites:
Navy - AS, EC
ser activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N300-8)

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